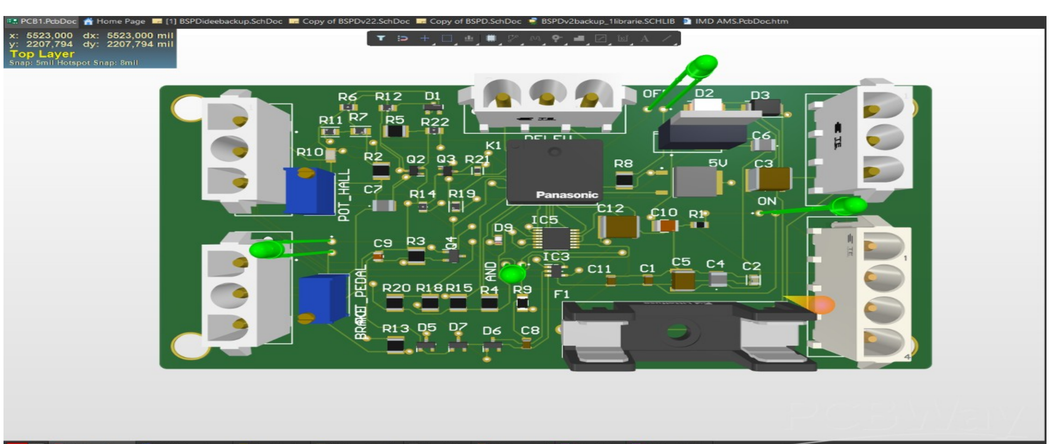
ASU LV Solo Mission Submission

1. Competition Rules For Safety Devices

- **<u>Brake System Plausibility Device (BSPD)</u>**: A safety circuit meant for detecting hard braking and continued acceleration simultaneously, its main function is to cut power and start the shutdown circuit (SDC) if prompted, to stop dangerous vehicle behavior and unnecessary strain and or overheating.

Key Notes:
1. Standalone non-programmable circuit. **(T11.6.1)**
2. Must open SDC when hard braking +
EV: ≥ 5kW power delivered to motors.
CV: throttle ≥ 25% over idle. **(T11.6.1)**
3. Must trip if implausibility persists ≥ 500ms. **(T11.6.2)**
4. Must be powered from LVMS. **(T11.6.3)**
5. No extra functionality, interface limited to bare minimum, Sensors/Power cant be routed to other devices. **(T11.6.4)**
6. Current sensing and brake pressure set at thresholds of 5kW and ≤ 30 bar respectively. **(T11.6.5 and T11.6.6)**
7. Each sensor signal wire must be disconnect-able for inspection, must classify them as system critical signals (SCS), Must demonstrate full functionality at tech inspection. **(T11.6.7-9)**
8. BSPD + sensors must not be installed inside the (TSAC). **(T11.6.10)**



- Insulation Monitoring Device (IMD): A device meant for checking electrical insulation between the tractive system (TS) and chassis, meant for monitoring the resistance between the two and triggering the SDC if the resistance drops below a certain threshold, to prevent leakage paths and electrical shocks.

Key Notes:
1. Every vehicle must have an IMD installed in the TS system. **(EV6.3.1)**
2. IMD must be a Bender A-ISOMETER iso-F1 IR155-3203/-3204/-4203/-4204, or Bender ISOMETER iso165C-1/iso175, or approved equivalent (criteria: vibration robustness, operating temp range, IP rating, direct output, self-test, not powered by monitored system). **(EV6.3.2)**
3. Response value must be set to ≥500 Ω/V, relative to max TS voltage. **(EV6.3.3)**
4. Response value must not be changed after electrical inspection. **(EV6.3.4)**
5. IMD must be connected on the vehicle side of the AIRs. **(EV6.3.5)**
6. One ground measurement line to grounded TSAC, other to main hoop; separate conductors rated ≥max TS voltage; open circuit in either must open the SDC. **(EV6.3.6)**
7. On insulation or IMD failure, must open SDC without programmable logic influence. **(EV6.3.7)**
8. Red “IMD” cockpit indicator, visible in bright sunlight, must light up when SDC opens and stay lit until manually reset; controlling signals are SCS. **(EV6.3.8)**

- Accumulator Isolation Relays (AIRs): Are normally-open mechanical relays located inside the Tractive System Accumulator Container (TSAC), meant for disconnecting both poles of the accumulator when the shutdown circuit is triggered. Their main function is to isolate the accumulator from the rest of the vehicle, ensuring that no tractive-system voltage remains outside the TSAC and reducing the risk of electrical shock or damage.

Key Notes:
1. At least two AIRs must be installed inside each Tractive System Accumulator Container (TSAC). **(EV5.6.1)**
2. The AIRs must disconnect both the positive and negative poles of the TS accumulator. **(EV5.6.2)**
3. When the AIRs are open, no tractive-system voltage may be present outside the TSAC. **(EV5.6.2)**
4. The vehicle side of the AIRs must be galvanically isolated from the accumulator side. **(EV5.6.2 and EV1.2.1)**
5. AIRs must be mechanical, normally-open relays. **(EV5.6.3)**
6. Solid-state relays are not permitted. **(EV5.6.3)**

- Brake Over-Travel Switch (BOTS): A mechanical safety switch designed to detect excessive brake-pedal travel, which can indicate a failure in one of the hydraulic brake circuits. Its main function is to open the shutdown circuit and stop the vehicle's power system from continuing to operate during a brake failure.

Key Notes:
1. Must be installed as part of the shutdown circuit. **(T6.2.1)**
2. Excessive brake-pedal travel must open the SDC if at least one brake circuit fails. **(T6.2.1)**
3. Must function at every brake-balance and pedal setting without damaging the vehicle. **(T6.2.1)**
4. Repeated activation must not close the SDC, and the driver must not be able to reset it. **(T6.2.2)**
5. Must be a mechanical single-pole, single-throw switch; multiple switches may be connected in series. **(T6.2.3)**

- Inertia Switch: A mechanical crash-detection device designed to recognize a strong impact and open the SDC. It helps stop electrical power and acceleration after a collision, reducing the risk of further movement, fire, or electrical damage.

Key Notes:
1. Must be included in the shutdown circuit and open it when an impact occurs. **(T11.5.1)**
2. Must remain latched after activation until it is manually reset. **(T11.5.1)**
3. Must trigger at an omnidirectional peak acceleration of ≤8 g for a half-sine pulse lasting ≥50 ms, or ≤13 g for a pulse lasting ≥20 ms. **(T11.5.2)**
4. Must not contain semiconductor components. **(T11.5.3)**
5. Must be rigidly mounted according to the manufacturer's instructions and removable for testing. **(T11.5.4)**

- Tractive System Active Light (TSAL): An external safety light used to clearly show whether dangerous voltage is present in the tractive system. It warns drivers, team members, marshals, and scrutineers whether the system is energized or safely isolated.

Key Notes:

1. The vehicle must have one dedicated TSAL that only indicates the condition of the tractive system. **(EV4.10.1)**
2. The red light must flash continuously at 2–5 Hz with a 50% duty cycle when dangerous TS voltage is present. **(EV4.10.2)**
3. Dangerous voltage means exceeding the lower of 60 VDC/50 VAC RMS or half the nominal TS voltage. **(EV4.10.2)**
4. The green light may remain continuously on only when all AIRs and the pre-charge relay are open and vehicle-side

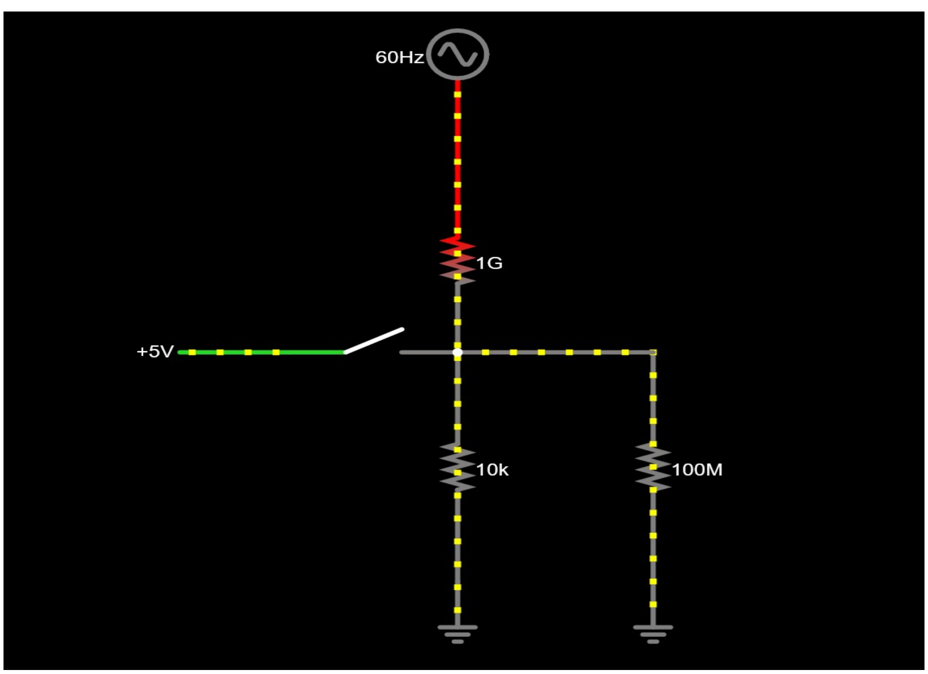
voltage is within the safe limit. **(EV4.10.3)**
5. The red and green detection circuits must operate independently. Both lights may be active simultaneously if the detected conditions require it. **(EV4.10.4–EV4.10.6)**
6. The TSAL must be positioned near the main hoop and remain clearly visible from around the vehicle, including in direct sunlight. **(EV4.10.7–EV4.10.8)**
7. The TSAL and its required detection circuits must use hard-wired electronics; software control is not permitted. **(EV4.10.9)**
8. A green cockpit indicator marked “TS off” must illuminate whenever the TSAL’s green light is active. **(EV4.10.10)**
9. The TSAL must remain powered for at least 15 minutes after the LVS is switched off. **(EV4.10.16)**

2. What is PCB crosstalk, and how is it reduced?

Crosstalk is unwanted coupling from an aggressor trace into a nearby victim trace through electric-field capacitance, magnetic-field mutual inductance, or shared return impedance. It becomes worse with fast edge rates, long parallel runs, small spacing, high impedance, and a broken return path.

Ways to reduce it:
- Increase spacing and minimize parallel length; route adjacent layers in different directions.
- Keep a continuous reference plane close to the signals so return current has a short path.
- Control impedance and terminate fast lines when required; slow the edge with a small series resistor if timing allows.
- Separate clocks/switching nodes from analog and high-impedance inputs; avoid routing across plane splits.
- Use differential routing, ground shielding/via fences, or extra layer separation when justified

3. Build this circuit in your simulator and answer the following questions:



a) What is this circuit's primary function? This circuit represents an active-high digital input circuit with a pull-down resistor. It could be used to connect a button, act as a limit switch, or as a digital sensor to a microcontroller. When the switch closes, the input receives +5 V and reads HIGH. When the switch opens, R1 pulls the input to ground so that it reads a stable LOW.

b) What do the components inside the dashed box represent? The 50Hz (60 in mine) AC source and R3 represent unwanted electrical interference from nearby wiring. The 50 Hz source represents the electric field produced by household electricity or other sources, while the very large 1 GΩ resistance represents the weak coupling between that interference and the input through air, insulation, or a body.

c) R1 — the main character

1. What happens if R1 is removed?

When R1 is removed and the switch is open, the input no longer has a strong connection to ground. In the simulation, the 50 Hz interference can cause the input voltage to fluctuate instead of remaining at a stable 0V. R2 still provides a very weak path to ground, but its 100 MΩ resistance is too large to provide strong noise protection. In a real IoT device, this could cause random readings, false button presses, or unexpected switching.

2. What is an input called when R1 is missing?

It is called a floating input. More precisely, the simulated input is very weakly biased through R2, but it behaves similarly to a floating input because of its extremely high impedance.

3. How should R1's resistance be selected?
A 10 kΩ resistor is commonly used because it provides a strong enough pull-down to keep the input voltage within the LOW threshold of most microcontrollers while still limiting current and power consumption when the switch is closed. However, the final value should be checked against the specific microcontroller's logic thresholds and input-leakage current.

d) What are the arrow and R2? The arrow points toward the current flowing into an input pin of a microcontroller, digital logic gate or an oscilloscope. R2 represents a load resistance where sensitivity is high.

e) Active signal changed from 5V to 0V. R1 should be moved from between the input and ground to between the input and +5 V, making it a pull-up resistor. The switch should then connect the input directly to ground. When the switch is open, R1 keeps the input at 5 V, or logic HIGH. When the switch closes, the input is connected to 0 V, producing logic LOW. Because the circuit becomes active when its input is LOW, this topology is called an active-low input with a pull-up resistor.

f) Bringing your hand close to the circuit. The human body acts like an antenna and couples nearby 50/60 Hz mains interference into the circuit through parasitic capacitance. Bringing your hand near an unshielded high-impedance input could therefore make its voltage fluctuate. If the input is floating or weakly biased, this interference could cause unstable readings, false button presses, or random switching between HIGH and LOW.

4. Decoupling capacitors vs bulk capacitors:

A **decoupling capacitor** is a small capacitor, commonly around 100 nF, placed as close as possible to an IC's power and ground pins. It supplies the IC with quick bursts of current and helps remove high-frequency noise from the power line. A **bulk capacitor** is much larger and is usually placed near the power input, voltage regulator, or a group of components. It stores more energy, helping prevent the supply voltage from dropping when the circuit suddenly requires more current.

5. Differential pairs and critical routing rules:

A **differential pair in a PCB** is a set of two complementary copper traces that carry equal and opposite signals to transmit data reliably while reducing noise. The traces should be routed close together with consistent spacing, matched lengths, the correct impedance, and the same reference plane. Unnecessary vias, sharp bends, and gaps in the reference plane should be avoided.

6. Polygon pours:

A **polygon pour** is a large area of copper on a PCB connected to a specific net, usually power or ground. It provides a low-resistance path for current, improves voltage stability, helps spread heat, and creates a better return path for signals. Proper clearances, grounding vias, and the removal of isolated copper areas should be considered when using it.

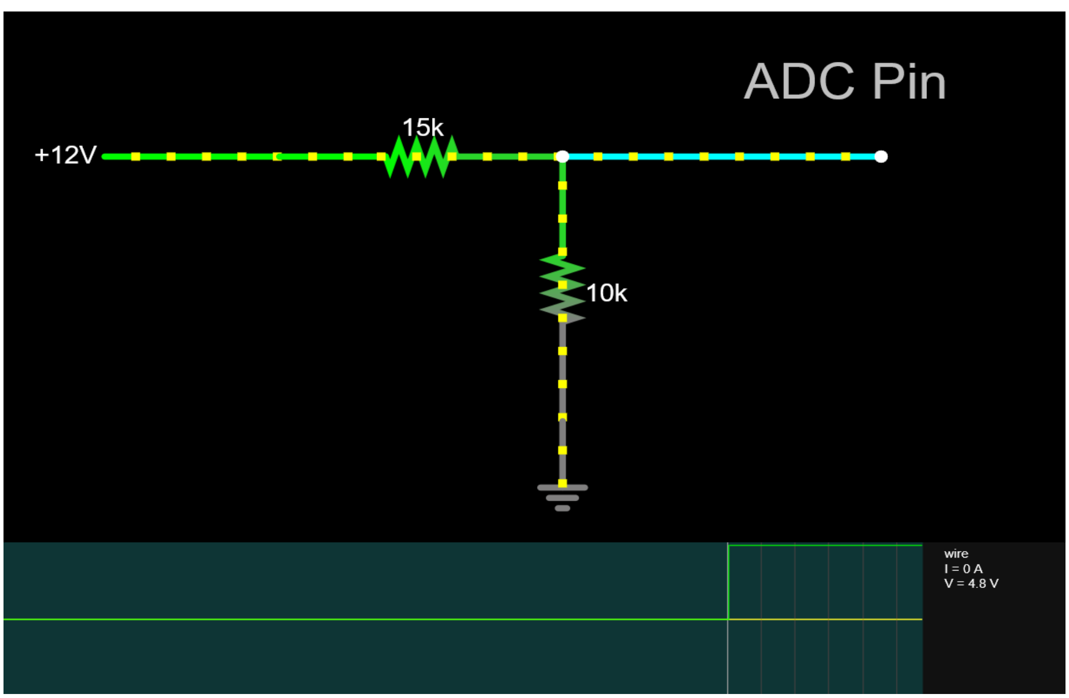
7. Ex:

Ill use a voltage divider made from two resistors: a 15 kΩ resistor between the sensor and ADC input, and a 10 kΩ resistor between the ADC input and ground. This reduces the sensor's 0–12 V output to approximately 0–5 V while consuming no more than 6 mW at 12 V. In a real circuit, resistor tolerances and the ADC's input requirements should also be considered to prevent the voltage from exceeding 5 V.

$$V_{ADC} = 12\left(\frac{10K}{15K + 10k}\right) = 4.8V$$

$$I = \frac{12}{25K} = 0.48mA$$

$$P = 12(0.48mA) = 5.76mW$$



8. Two interrupts arrive at the same time: If two interrupts occur at the same time, the processor handles the one with the higher priority first while the other remains pending. In ARM Cortex-M microcontrollers, a lower priority number means higher priority. If both interrupts have the same priority, the NVIC uses their interrupt number to decide which runs first, then handles the other afterward.

9. Reading BNO055 Euler angles over I2C: To read the Euler angles, place the BNO055 in NDOF fusion mode and use I2C address `0x29` or `0x28`, depending on the COM3 pin. Read six bytes from registers `0x1A–0x1F`, which contain heading, roll, and pitch. Combine each pair of bytes into a signed 16-bit value, then divide by `16` to convert the readings into degrees. Reference: Bosch BNO055 Datasheet, Section 3.6.5.4, page 37; Section 4.2.1, pages 55–57; Section 4.3.61, page 76; and Section 4.6, pages 100–102.

10. Communication protocols and examples: A communication protocol is a set of rules that allows electronic devices to exchange data correctly. Examples include UART, which uses TX and RX lines; I2C, which connects multiple devices using two wires; SPI, which provides fast communication using separate data, clock, and select lines; and CAN, which provides reliable communication between multiple devices and is commonly used in vehicles.

11. STM32F4 vs Raspberry Pi Zero vs Atmega:

- STM32F4: A powerful 32-bit microcontroller family designed for fast and reliable real-time control. It is commonly used in motor control, data acquisition, automotive systems, robotics, and advanced embedded applications.

- Raspberry Pi Zero: A small single-board computer capable of running Linux. It is used for applications requiring networking, file storage, cameras, graphical interfaces, or programs written in languages such as Python.

- ATmega: A simple and low-cost family of 8-bit microcontrollers, commonly found in Arduino boards. It is suitable for basic sensor reading, LED control, simple automation, and beginner embedded projects.

Sources and References (GIT LINK AT THE END)

Click any blue source title to open the original reference.

~~1. Formula Student Rules 2025, Version 1.1~~

Questions 1: BSPD, IMD, AIRs, BOTS, inertia switch, and TSAL requirements.

~~2. Texas Instruments - High-Speed Layout Guidelines (SCAA082A)~~

Questions 2, 4, and 5: crosstalk, bypassing, return paths, and differential routing.

~~3. Texas Instruments - PCB Layout Design Guidelines to Meet CISPR 32 (SLLA561)~~

Question 4: placement and roles of local decoupling and bulk capacitors.

~~4. Microchip - ATmega328P Complete Datasheet~~

Question 3: defined input levels, floating inputs, pull resistors, and logic thresholds.

~~5. STMicroelectronics - STM32F103x8/xB Datasheet~~

Questions 3 and 7: digital thresholds, leakage, ADC limits, and electrical ratings.

~~6. Texas Instruments - Understanding 50/60 Hz Human-Body Common-Mode Pickup~~

Question 3(f): capacitive mains coupling through the human body near high-impedance inputs.

~~7. Falstad CircuitJS1 Electronic Circuit Simulator~~

Question 3: simulator used to examine the pull-down input and interference model.

8. Altium Designer - Polygons on Signal Layers

Question 6: polygon pours, net assignment, clearances, dead copper, and connection styles.

9. STMicroelectronics - How to Optimize ADC Accuracy in STM32 MCUs (AN2834)

Question 7: divider source resistance, ADC sampling time, and input capacitance.

Sources and References (continued)

Click any blue source title to open the original reference.

10. STMicroelectronics - Cortex-M3 Programming Manual (PM0056)
Question 8: NVIC priorities, pending interrupts, preemption, and tie-breaking.

11. Bosch Sensortec - BNO055 Datasheet, Revision 1.8
Question 9: I2C address, Euler registers, byte order, scale, and operating mode.

12. NXP Semiconductors - I2C-bus Specification and User Manual (UM10204)
Question 10: I2C wiring, addressing, open-drain operation, timing, and arbitration.

13. STMicroelectronics - STM32F1 Reference Manual (RM0008)
Questions 7 and 10: ADC, USART/UART, SPI, I2C, and CAN peripheral operation.

14. STMicroelectronics - STM32F4 High-Performance MCU Series
Question 11: STM32F4 architecture, performance, peripherals, and embedded use cases.

15. Raspberry Pi - Raspberry Pi Zero Product Page
Question 11: processor, memory, operating system, connectors, and SBC use cases.

16. Microchip - ATmega328P Product and Documentation Page
Question 11: 8-bit AVR features, memory, peripherals, voltage range, and typical uses.

17. ASU LV Mission - GitHub Repository
Project files, simulation material, and supporting work for this submission.